

Jean-François Cloutier: “An experiment in artificial agency”

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all right that was a great presentation this will be the last of the recorded presentations for the Symposium and last video on this part one stream we have JF clae an experiment in artificial agency an experiment in artificial agency I'm Je I'm a research fellow at the Active infant Institute and I do research in cognitive robotics one of principal question I ask myself is what does it take at a minimum for an autonomous robot to learn to survive in a world it knows initially almost nothing about other questions which follow is are can a robot act for its own reasons does it have agency can its agency be programmed and would such a robot be an active inference agent the one way to find out is to actually build an artificial agent uh and by artificial agent I mean a stance making adaptive and self-directed robot so this project is a continuation of of a personal project that I started quite a few years ago and 3 years ago here's where I was I had a Lego robot uh with uh quite a few sensors um like ultrasonic uh sensors infrared sensor um a color sensor luminum sensor um Motors of different types all this uh running uh from a Raspberry Pi with a board that provides an interface between the Raspberry Pi and the various Lego motors and sensors and the robot also had a 360 degree Beacon that allowed one robot to see another through uh the other robot's 360 degree Beacon Seeker so let's look at them in action I named them uh Carl and Andy for I think quite obvious reasons the Carl had gone through quite a lot of training runs so it was better at finding the food which is the piece of paper on the flo um with a a beacon infer Beacon providing the scent the attractive scent we can see that Andy is uh having a bit of a hard time and they see each other so that if one robot starts acting panicky which they do if they Collide too often the other will notice the pattern of movement and if one robot makes a beline then the other robot will perceive that and say aha that's where the food is and try to move in the same direction but we see them uh moving about and trying to get a food and uh eating the food and Pa is clearly uh more apt at this task than andd and what they've learned through various trials are the policies that um are most successful at finding the food so that's where I was um three years ago the

way I implemented uh the cognitive capabilities quote unquote of these robots was based on a uh architectural model uh described by Marvin Minsky in his book The Society of Mind this um architecture posits that interesting behaviors emerge from many simple actors that interact in simple ways so we're looking at cognition as a collective intelligence the implementation was a hierarchy of these simple processes I call them cognition actors and they formed an abstraction hierarchy with at the very bottom um cognition actors that simply wrap sensors in uh effectors Motors and then at the very bottom we have cognition actors that care about very simple things like the location of food then at a higher level information obtained from the food location cognition actor feeds into the food approach cognition actor and this one is in tries to figure out am I getting closer to the food and so forth and so on other cognition actors uh are concerned with avoiding collisions and avoiding dark places now each cognition actor as I said is a separate process and each such process uh does very simple things it makes predictions about what it will observe next it then updates its beliefs based on uh predictions that were successful or prediction errors received and then selects a policy randomly in the beginning but then based on on uh what was successful in the past and applies that policy acts and then decides and then predicts what and what the next incoming Sensations will be and so forth and so on essentially implementing an UDA Loop observe Orient decide and act there's one big problem though that this Society of Mind these this hard of cognition actors is predefined it's hardcoded for each robot it's the same the only thing that's learned is what policy is more effective over time based on experience so the robot acted for my reasons for the programmer's reasons not its own the robots had autonomy but no agency in 2022 I joined the active inference Institute and continued this project under the uh symbolic cognitive robotics um tag and I asked myself what if what if a robot started with only a rudimentary Society of Mind what if it didn't have that pre-built Society of Mind this this hierarchy of function actors it started only with the the the most most basic ones the ones that wrap sensors and Motors then the Society of Mind uh would grow from experiencing autonomous engagement so the robot would start and with all these very basic um prior cognition actors for sensing and affecting and then would create new cognition actors then these cognition actors uh would be very uh um very simple in the sense that they would take their Sensations from the the uh uh the sensors themselves and would act uh by uh engaging the motors and then uh observe how it changes its its its perceptions of say color or distance and whatnot Etc so action actors would be added to the Society of Mind bottom up to make sense of observations and actions now each new cognition actor would select an umelt composed of cognition actors one

level below and how do we know that a a a particular level of cognition actors in This growing hierarchy is complete it's when all con actors one level below uh belong to anelt now it would not only grow but it would also shrink because whenever uh Cog actors were recognized to be hopelessly ineffective they would eventually be removed and new cognition actors would be would take their place and the hope would be that uh as this cognition hierarchy grows uh becomes more uh competent we would have a robot that starts acting uh intelligently and and in survivable ways so the Society of Mind would grow shrink and evolve as needed to survive now that's that's a key Point here um this is an implementation of um a form of mortal Computing obviously the the robot is physically the we Wheels uh the sensors themselves uh are not mortal they're they're like a a a computer on Wheels you turn it off you turn it on it's it's still alive it's still it hasn't died by from having been turned off however the Society of Mind itself is Mortal it grows it evolves in such that the robot will be competent at finding the food it needs to keep the society of Mind alive so what we have here is a um Immortal Society of life of of mind trying to um achieve a level sufficient level of Competency to maintain itself this is work in progress this is a um a system that I'm in the process of building and it's composed of these subsystems agency where uh the smart are for growing shaping uh and evolving a society of mind there's another subsystem that it runs on the uh robot itself which I call Body which provides a um an interface to uh activate uh Motors and to read sensors there's also a a a version of the body subsystem which will run in a simulated world and the subsystem world implements that simulated environment it will also um animate an a real environment and then there's a subsystem that I call Observer which provides me or any uh other human being with a a window into what's going on with the within the Society of Mind of um the robot the agency subsystem runs on a powerful computer the the body subsystem the physical one runs on the brick pie which is powerful enough to run the uh significantly complex programs but not too computationally intensive ones and um world and observer and and body simulated run on a um a separate computer the Karma body subsystem as I said gives access to real and virtual sensors and effectors and here I have all the um the sensors I want to use and the motors um removed from a a the robot just like connected directly to the uh Rick by and they are uh they can be uh activated they can be uh reached uh via an um arrest API which is essentially um they exposed a a a web interface uh to the agency subsystem the karma world subsystem uh here we see the simulated environment it's a grid World um the the gray tiles represent just the the surface and with a brown tiles representing obstacles so the the robot will bump into these obstacles and will perceive them with its uh

simulated distance sensors the green patch represents a Food Patch and the robot needs to find food in order to consume the food and replenish its U its resources if it runs out then it dies the uh we can see that some Tiles are are uh darker others lighter and they this gradient this luminous gradient corresponds to um um a a scent gradient so the closer the robot is to the food the lighter the tiles are there will also be um poison patches which will be red so detectable as a a red surface on on the simulated floor the robot must learn to avoid poison food because that will harm it and if it's armed uh if it's um excessively harmed then it will also die when a robot spends a enough time over the food the food is consumed and uh another Food Patch appears somewhere else so the the environment is dynamic and it changes with the uh with the its interactions with the robot that's an important uh element here now the um the karma agency subsystem is the most complex and and the the key subsystem and it implements a um generative cognitive architecture uh for the robot implementing agency so we're going to it's going to implement um cognition actors as we've mentioned already it will Implement something I call metacognition actors a very limited uh understanding of the the concept of metacognition one metacognition actor per level in the hierarchy and the cognition actors are responsible for adding removing uh cognition actors as uh needed as required then very very importantly we'll have well-being actors and these actors are represent the um homeostatic values that the society mind must keep in and within a acceptable bounds in order to survive we have fullness integrity and engagement we'll talk in more details about what they are later and then very importantly uh there's an app perception engine actor uh last year at the Symposium I presented the um my implementation of an a perception engine and this is the um very important part of component of the Society of Mind it is responsible for making sense of observations in order to be able to make predictions we'll go into more details as well now let's go and look at these well-being actors they Implement uh homeostatic uh States and I have three one is fullness which represents how much energy the uh the agent as a whole the robot has available to do things that could be Computing doing some comp uh expensive computation or um just moving about so there's a an energy budget that gets depleted and also gets replenished when the robot finds food and spends enough time in a Food Patch there's also the Integrity well-being actor and that represents uh the health of the uh agent the if the agent uh the robot suffers too many collisions then Integrity goes down and integrity is is recovered over time think think the think about lives in in a video game that's pretty much the same concept and finally a little bit bit more abstract uh well-being actor is engagement and it represents how to what degree the robot is engaging its environment we

want to avoid the dark room situation where the robot uh stops doing anything uh stops moving stop engaging its environment it h it it has a it it it feels quote unquote an imperative to to engage its environment now the agent as a whole meaning the Society of Mind of the agent strives to maximize well-being for the entire Society of mind now low well-being levels will signal risk to the survival of the Society of Mind so that information will be available to each individual cognition actor and to the metacognition actors and will modify uh constrained uh constraint what they do key concept here is that well-being imparts normativity to beliefs beliefs acquired in the context of a um low fullness or low Integrity becomes a uh unpleasant undesirable belief now this this this normativity Associated to beliefs will focus attention and motivate action as we'll see later on so well-being actors aggregate metrics uh received from the various cognition actors and then aggregate these metrics and broadcast uh wellness events back to the cognition actors creating a give essentially informing them of the um ostatic State uh of the agent as a whole now these metrix include the work that's been done the food that's been consumed these these two will impact the fullness metrics and values then uh poison consume Collision count will impact the Integrity overall Integrity of the agent and then uh po predictions made predictions received prediction errors sent policies executed we'll see that in more detail later all these are indicative of the uh Society of Mind engaging its environment now metal cognition actors their role is pretty straightforward they're responsible for adding cognition actors that are missing and to remove underperforming cognition actors each cognition actor is assigned one level in the abstraction of hierarchy and it looks at the conion actors under its supervisions supervision and and determines if all the conion actors below them are part of one or the other's umelt so the notion of umelt here a cognition actor can only observe or act within its umelt and what is the umelt of a cognition actor it's other cognition actors of a lower level of abstraction so at the the first level of conion actors the prior level uh where we have the um where they wrap sensors and and Motors um they don't they they don't have a nelt well they do it's the actual external environment of the robot they don't have an etal enel anything above this level uh the cognition actors have as part of their umelt other cognition actors and they constitute essentially the internal umelt of of within the Society of Mind so if a a one layer of the um Society of Mind is complete it covers all all the cognition actors below then it's time to create a cognition actor at the level above create a new layer of the Society of Mind and when you create a cognition actor you give it or the metacognition actor gives it its umelt mostly randomly with um the constraint that there's going to be overlap between umelt of various

cognition actors and making sure that they have access uh to the mots directly or indirectly now when when well-being is low within the Society of mine when the well-being actors are reporting low wellbeing levels this might accelerate the removal of cognition actors the metacognition actors being aware that we're not doing well now will say okay which cognition actors are underperforming our expectations and let's remove them in in order to for them to be replaced with new C actors which hopefully will do better and if uh well-being is low this will also slow down adding new cognition actors so this is an influence of well-being on the the actions of the Mion actors I want to say adding and removing um it's important to uh to to understand that adding is bottom up and removing is top down for example if I have a a meta if I have a cognition actor that has let's say two other conion actors in itsv I cannot remove a con actor in itsv because it depends depends on it but I can remove a cognition actor that is not in any other's umelt so that removal will be from the top to bottom and addition will be from the bottom to top what does a cognition actor do well a cognition actor observes its umelt of other cognition actors it generates beliefs from analyzing its observations and then it generates policies in order to modify or validate its beliefs and it does this this observe believe and act always within itself and one time frame at a time so each cognition actor goes through its more or less UDA Loop one time frame at a time now the the lower the lower a cognition actor is in the uh Society of Mind the more concrete it is the less abstract it is um the shorter this time frame will be it will cycle faster it's important also to realize that there there's no there's um there's no synchronization between cognition actors each starts its time frame independently of other cognition actors and so that the the they may overlap but they don't coincide they don't overlap completely and the more abstract the cognition actor is the higher up it is in the hierarchy of cognition actors within the Society of Mind the longer its time frame will be which means that very abstract cognition actors will update their beliefs more slowly and will take actions uh not as quickly as a a more concrete low-level cognition actor so we can see here the the cognition actor at the large one surrounded by its lowlevel cognition actors the what does it observe what does the con cognition actor observe in its unel well it observes it observes the belief beliefs of its of the cognition actors in itsv so as a more abstract cognition actor I observe the beliefs of the more concrete cognition actors in my umelt then based on these beliefs and as we'll see based on the normativity associated to the beliefs they good beliefs or they bad beliefs the cognition actor will uh um decide on what its goals are in this current time frame the goals would be modifying a particular belief or validating a particular belief then it will generate policies to attain this goal and then emit this policy to its envel and

say okay do what I want and we'll see in more details what do what I want means So within each time frame a cognition actor Will predict its umelt after having made sense of it that's a key uh aspect we'll we'll go into more details it will update its beliefs by analyzing past and present observations so a a cognition actor has a memory of the current observations and this the current time frame and observations made in previous time frames and it will update its beliefs but from analyzing current observations and past observations then it will select a goal affecting a belief in it tends to impact and construct a policy and recruit itself to realize it so if you look at the little U redesigned UDA Loop here first there's predict make predictions about beliefs of other COG actors in uh my umelt in order to predict I need to have a causal model of my observations that's what a perception will do we'll see in more details then the as a cognition actor update my beliefs select goals act and back to predicting in the next time frame a perception is defined here as the process of assimilating sensory information into to a coherent unified pole so let's look at uh predicting what are the observations of a cognition actor observations are predictions it makes about the beliefs of other cognition actors which are in its umelt plus that plus prediction errors so it makes predictions of itself of ition actors in its umelt about their beliefs and the umelt cognition actors receive that prediction look at it does it corresponds to my current beliefs if yes not saying anything I'm not contesting the prediction if it doesn't match then I emit a prediction error and the cognition actor accumulates these predictions that are uncontested and the prediction errors and together they constitute the Cog actor's current observations so we'll see that not all observations receive the same attention that they're not subject they're not equally subject to prediction there's a prioritization there's a notion of attention we'll see that a bit later attention prioritizes which observations to update so the cognition actor starts a time frame with observ observations made in previous time frame and if there and it wants to make to make those observations current to the current time frame as many of them possible but there's a limited amount of time in which to um update these observations so it the C actor will need to prioritize which observations from the previous time frame to update into the more current observations so which one will it try to update first well it will look at the predictions that were made by a higher level predic cognition actors about its beliefs and knowing which obs a support the beliefs will say ah these is are the more these are the observations that um more uh abstract more important cognition actors are making about me my state my beliefs so I'm going to I'm going to try to make sure that I have upto-date observations that are of interest to my parent my my parent cognition actors so that's the first one I care about the first observations I care

about the ones that are of interest to my parent function actors actor sorry or actors the second most important observations in in terms of uh updates are the observations that support the strongest desirable and undesirable beliefs remember that uh beliefs are the result of analyzing observations current and past if I have very strong desirable or very strong undesirable beliefs then I will want to verify that those OB observations are still current so there will be the next uh observations that I will try to uh update then finally the the observations that uh support weak or neutral beliefs and then whatever remaining observations they are so the conition actor will take will prioritize the observations it it that existed from the previous time frame and say okay I'm going to now make predictions um that these observations are still valid or not valid and if those there are obser prediction errors then then the the the observations are updated to the prediction error and if the observation is not contested then the observation is is is um the one that is current this implements a predictive processing uh architecture because as you can imagine a high level cognition actor will make predictions about beliefs uh of the the the its children actors this will trigger predictions about the observation supporting dis beliefs one level down to the cognition actors in the layer below and all the way down so predictions flow down and prediction errors blow up through the entire hierarchy of uh the conion actors in The Society of mind now in order to make predictions we need to have made sense of uh observations that's what app perception does so we need to make sense of observation in order to predict them now in order to to explain the concept of our perception in which in last year's presentation I went into quite a lot of details but just I want to give a an overall understanding here let's take the example of observing uh two lights light a and light B and the light can either be on or off and at different steps in time the lights may change their state from on to off off to on or not so here we have a set of observations of light A and B over time and sometimes we may not be able to observe a light like uh light a in in the fourth state is a gray rectangle we don't know but we have a good enough set of observations and now the goal of the app Apperception engine is to find a causal theory that explains how the observations change over time why are we seeing some light a turn on after having been off or Etc in in order to formulate a causal Theory kind of a little program that when run recreates what we've observed the Apperception engine may need to imagine another light it might need to imagine some kind of connection between the lights so the Apperception engine will in in formulating a causal Theory will do some abduction some imagining of unobserved latent objects so we have a number of observations that are remembered from previous time frames we may uh we we don't accumulate all observations forever we forget the older ones and then we feed these

observations into the app perception engine out of that comes a little program a causal theory that when when applied to the latest observation will produce the next incoming expected observations so that's a way of of making predictions of incoming uh observations from current and past observations from the current observations given a causal theory so you feed the latest observation you feed all the observations into the app perception engine it gives you a causal Theory you feed the latest observation into the causal Theory as input the output is incoming predicted observation so an Apperception Apperception means finding a causal explains by all the rules of the theory and predicts so a actor accumulates observations wants to make sense of them in order to predict incoming observations and for other reasons as well which we will see and it will what it does is okay it gives the app perception engine actor its observations over time and says please explain to me what I'm observing give me a causal Theory so I can predict future observations given this causal theory if one is found successfully then the contion actor can predict incoming observations and what does it observe it observes the beliefs of cognition actors in itself more more concrete cognition actors and then if the prediction is correct if the causal theory is correct and accurate we don't get a prediction error if the causal theory is partially correct some times correct then we get a prediction error so getting prediction errors undermine confidence in the causal theory that the Kung actor has received from the aips and engine to explain what it sees in its umelt so it's quite possible that there's a moment in time when the observations when the causal theory is is quite effective and it's accurate but then something changes in in the state of theel function actors such that the causal theory is is missing is is very hit and miss and it may come a point where the contion actor says I don't like that causal Theory anymore it's it's producing too many prediction errors so it's going to ask the perception engine actor give me a new causal Theory based on the latest observations that I've accumulated In My Memory Okay so we looked at um how cognition actors predict incoming observations in order to uh accumulate those new observations to update those new observations based on whether or not these uh predict the predictions uh are contested by prediction errors or not so now once we have a new set of current uh observations within this time frame that we are operating in time to update beliefs beliefs are if you want um integrate are are the integration of observations into a a a more aggregated um analytical uh assertion so cognition actor updates its belief from analyzing remembered observations and also uh remembered actions that it that that that it took so and we'll see what kind of of integration of observations and Analysis of observations lead to what type of beliefs but whatever type of belief it is a belief will be desirable or

undesirable depending on how it correlates with the agent's well-being if a belief is being held mostly while the agent as a whole is feeling bad it has low resources or low Integrity or or limited engagement then these these beliefs are going to be tainted by the uh well-being um levels of the agent at the time those beliefs are held now that's what makes a belief desirable or undesirable what makes a belief strongly held is if it is supported by observations those observations from which analysis the belief is derived if it is supported by observations that are predicted by an accurate causal Theory we have an excellent causal Theory it's it's it rarely misses so um the the cognition actor is feeling confident about its observations and it makes the the whatever belief is is produced from analysis of these successfully predicted observations better supported more strongly held now if a belief persists over many time frames thanks to or in spite of taken action taken to validate or invalidate it if the belief persists then it's even more strongly held it's it's it's there's more there's increased confidence that the belief is uh represents uh what's really going on in the umelt of the cognition actor now there are different kinds of beliefs beliefs are are are generated are produced from the analysis of uh current and past observations and okay so there's abduction one type of beliefs is ISU abduction and it's a direct consequence of um having requested a causal Theory and using a causal Theory remember in in a causal Theory May um imagine a a latent object or a latent relationship between objects well that's that's a form of belief I believe there's an object that I don't see but must exist in order for me for for a causal Theory to make sense of actual observations I made so an abduction belief would be like I assume there's a hidden thing next to the one I see another kind of belief is just counting I'm I'm I'm next to two things I have made two observations that represent uh say being next to something and so I have two of them so I'm I can have a belief that I'm next to two things account belief now a trend belief is a it's a belief that represents the the the um discovery of a trend in a c certain observations over time for example example if um the con actor observes distances to some obstacle and those distances are getting smaller um over time then that's a trend my distance to a thing is getting smaller and then there's another kind of belief is is some observation ending stopping by getting closer to a thing stop so that's an ending kind of belief and then if the conion actor has uh taken some action well there's the belief that it tried to do something like I I tried to stop my distance to a thing from getting smaller so if the action was trying to end the belief that the belief in a trend I you can see now that beliefs are are are composed they're getting more and more abstract so I tried to stop my distance to a thing from getting smaller is a kind of another kind of belief it's an attempt kind of belief so we have all these different kinds of beliefs and

when a Cogan actor synthesizes the belief from analyzing observations it assigns of belief a globally unique predicate name parent cognition actors the cognition actors in which inelt this T belongs um the parent cognition actors observe predict the beliefs of cognition actors in in in their owns and reference them by name so if the there's a belief in umel K actor that some Trend exists the um we're getting closer to an obstacle well the this belief will be exposed will be made available for or observation to the parent as a named uh predicate why and how that belief uh came about what the observations are by the umel kin actor that led to that belief it's it's it's opaque there's a name and that's the name that's available for observation there's an a statement a predicate name predicate that's available for observation by the parent cognition actor only the cognition actor that holds the belief it named knows which of its own observations support the belief and how so if I say to you I believe I I believe getting closer that's something that can you can observe as a parent con actor ah my child con actor believes in getting closer but the parent cognition actor doesn't know why I believe we're getting closer doesn't have access to the set of observations that led to me holding the belief that we are getting closer all these distance observations that led to me having holding the belief that getting closer all that parent knows is that I hold the belief called getting closer and it's able to observe it observe it you know ending or or starting Etc but that's that's that's a key um a key concept that beliefs are are essentially named um there are predicates their names if you wish that uh hide the C the reason why the belief is held to parent contion actors that observe my beliefs so there's information hiding going on here so a cognition actor observes the beliefs of its children cognition actors and these these con these these beliefs that are observed are the consequences of the themselves observations about from of of cons actors one level below so my belief is is another if I'm a cognition actor my belief is another cognition actor's observation so it's really beliefs it's it's beliefs about beliefs about beliefs so it's beliefs all the way down that's that's a maybe a little tricky concept if I'm at the top as a cognition actor I observe beliefs of cognition s under me these beliefs of connection actors under me are supported by their observations of beliefs of cognition cognition actors under them so beliefs about beliefs about beliefs time to open a little parenthesis here maybe take a break the implementation that I'm working on of agency is a symbolic implementation so it's its agency realized from symbolic reasoning all the various agency actors the the cognition actors the mid cognition actors the well-being actors the uh the perception engine actors all run prologue programs prologue is a programming language and those prologue programs do symbolic infome the cognition actor as we've seen invents symbols invents gives names to beliefs it

discovers from analyzing observation so it invents symbols and it synthesizes these beliefs as symbolic statements getting closer to XYZ the app perception engine as we've seen very kind of briefly its job is to discover causal models and these causal models are symbolic programs they're little logic programs so agency is built on symbolic reasoning in my implementation closing the parenthesis now let's talk about precision and belief strength the more accurate a causal Theory the more precise it are are are considered its predictions so we saw earlier now precise predictions if they're not contradicted by errors do strengthen the beliefs the predictions observations support so you have a bunch of observations held by the actor observations in the moment remembered observations these observations are the result of predictions and or prediction errors and we analyze these these these observations coming from predictions to generate beliefs I'm getting closer to an obstacle from observations of distances now if my observations of distances were predicted by a really top-notch causal Theory one that almost never misses then I have confidence in the beliefs that have derived from my predictions I feel confident that I am indeed getting closer to an obstacle given all the the the distance observations that I've accumulated because I've accumulated them by making predictions using a very accurate empirically accurate causal Theory so precise predictions strengthens the belief that are derived from these from observations from predicted observations now the stronger beliefs are because I have may have strongly held beliefs and weakly held beliefs the strongly the more strongly held beliefs are the more attention they will receive When selecting when this the cogent actor needs to wants to select one and say I don't like this belief It's associated with poor well-being and I want to I want to change that belief I want I want to impact it I want to I want to get rid of that belief I want to make that belief not true or if it's a very pleasant belief oh I want to make sure that it is I want to validate it I want to keep that belief uh I want to keep holding that belief because it's a good one but I care whether this the belief is strongly held or weakly held because strongly held belief will get more attention will be the ones that'll be selected for validation or invalidation above above the the the weakly held beliefs so how do we impact beliefs that's the next step in the ud next two steps in the ud Loop of our cognition actor within each time frame it predicted based on its understanding of past observations it updated its belief from uh from analyzing predicted predictions that were realized or or error s predictions I those observations it updated its beliefs and now it says it needs to select a goal or may need to select a goal that say I want to get rid of this belief or I want to validate this belief and how am I going to accomplish this goal I'm going to select the policy and then and then ask my unal to act to realize this policy so that's what we're

looking at now so a cognition actor acts via policies it formulates a cognition actor strives to invalidate unpleasant beliefs and to validate Pleasant beliefs but it will prioritize strongly held beliefs whether unpleasant or or Pleasant a cognition actor will impact the belief validate or invalidated it holds by uh promoting or inhibiting the observations that led to it now I have a belief that we are getting closer to an obstacle and somehow it's it's it's an unpleasant belief maybe uh It's associated with in the past with um uh a decrease in in in U Integrity leads to bumping into things so I want to I want to put an end to getting closer to an obstacle I want to I don't want this belief that I'm getting closer to an obstacle to go away now I hold this belief because I had observations of reducing distances over time to some obstacle so I want to change these observations basically I want to I want to stop getting close having um having uh distances that get smaller I want to stop this so my observations of distances happen to be beliefs of PR actors in my umelt so what I'm going to do I'm going to say okay I'm going to select a number of of observations that support the belief I want to get rid of and I'm going to formulate a policy as a list of goals to my inelt CA I'm going to say I want you to get rid of the fact that I have small distances to an obstacle that's that's that's your goal so I'm going to I'm going to set a number of goals me the actors who believes we're getting closer to something and it's not doesn't feel good I'm going to tell I'm going to formulate a policy so that my umelt con actors will try to um impact their own beliefs that we're we have a small distance to an obstacle for example so that's that's a little bit subtle here so I as a cognition actor I want to modify a belief that belief is based on observations these observations are beliefs of umelt cognition actors so I'm going to say to myel actors I want you to to modify your own beliefs that I've observed I want you to change your beliefs beliefs that I'm observing that lead me to believe that I'm getting closer to an obstacle so I'm setting up a bunch of goals for my umelt cion actors so that if they realize if they if they modify the these beliefs then it will modify indirectly my own belief that I'm trying to get rid of okay so I'm basically telling my my umelt these are the goals I want you to realize so that I it I would indirectly realize my own goal Okay so this Pol a policy that I put together are a set of goals for my umel cognition actors to realize now a nelt conition actor will then given the goal it received formulates its own policy because we're I'm asking con actor in myel to modify some some belief or other okay says the con actor then I will formulate my own policies to do that to to achieve the goal that you that my parent cognition actor sent to me so a cognition actor will receive goals from parent con actors and these goals received from IR up will have higher priority that whatever goals the cognition actor might want to formulate on its own because the the umel function actor has its

beliefs has belief that it wants to validate has belief it wants to to to invalidate and left on its own to its own devices will do just that but if it receives goals from a parent con actor these take priority and it it will put aside its own concerns to uh address the concerns of the parent cognition actor now a cog actor cannot execute only one policy at a time which is why it will prioritize the ones that it receives from its parent and when all is said and done the cognition actor that initiated the policy will determine will will verify if this was successful or failure over time if the belief they wanted to impact was indeed impacted and a policy policy that's known to work will be reuse that's how con actors develop habits okay so where does the causal theory that the con actor uh holds guide the formulation of its policies because there was causal causal causal Theory serves two roles one is to predict incoming observations and the other role is to formulate an effective policy so let's say that cognition actor is a belief that wants to impact because uh [Music] the well-being me levels are are are low Associated to that belief so he wants to modify that that belief get rid of it well it needs to formulate a candidate policy which of the observations that I made that I've analyzed into this belief which which observations that support this belief I want to modify so which policy do I want to formulate well well what it will do is we'll say okay what do I need to do given what my what I'm observing right now so that I will observe something differently that will either remove the belief or or reduce it so it will use the causal Theory so what would happen if I did X if I I I I I if I asked my umelt to do XYZ so the causal Theory will answer that question we say well plug that into the CLE Theory turn the crank and this is what you would be uh be observing next if you were doing this if you if if you executed this action this is what you would observe next and then given what you would observe next you can calculate ah this these will be the anticipated beliefs out of these anticipated observations might the policy work with with the belief that I want to impact be impacted the way I want to impact it if so yeah let's use that policy otherwise let's look for another one so the cognition actor searches for an effective policy given its understanding of how things work in its umelt its causal Theory and and then applies the causal the theory predicts what would happen if it took this action if it if it requested this goal to be realized looks at the the the the resulting anticipated beliefs and decides was this a a a good effective policy and then and then it would admit that policy and ask it umelt please achieve these goals so that I achieve my goal so and and that percolates down because each cognition actor being given a goal from the parent Cog actor says ha how should I best realize that goal what policy should I formulate for my own child my own umelt in order to realize the goal that was given to me by my parent so it's policies all the way down

and it bottoms out when at the level where uh effects are produced moving Motors turning moving forward moving backward whatever okay so that's a lot of uh a lot of content describing this cognitive architecture this generative cognitive architecture what I hope will happen is through this growing shrinking evolving um um refining of this Society of Mind composed of of cognition actors um with uh in the context of of of of homeostatic measures provided by well-being actors I'm I'm hoping that with policies going down beliefs percolating up predictions going down predictions going up all this this this this changes and evolution of the Society of Mind all these all these activities from all of this would emerge agency and I I'm looking at that as as a form of operational closure with uh constraints flowing down constraints flowing up and eventually hopefully the um the Society of mine will explore a space of possible organizations and and and and and and space of of possible uh ways of defining cognition actors relating cognition actors together such that it leads to behaviors of the robot that preserves the life of the Society of Mind that Society of Mind never reaches a point where it runs out of fuel where it is damaged Beyond repair that's that's the overall goal I think of the Society of Mind as kinda autopoietic with all these um elements of the Society of Mind leading to other elements for example a causal Theory will lead to predictions predictions will lead to prediction errors prediction errors will lead to observations uh observation will be fed into a perceptions which will lead to an updated causal Theory and and observations lead to to beliefs um beliefs lead to policy um a CLE Theory will determine which policy is more likely to work policy leads to effect effects lead to New Sensations Sensations lead to new observations and so forth and so on so this I look at this as a a kind of a um a a set of of of processes that support each other and in in something of a of a closed set of complex Loops leading to a Society of Mind that survives over time now there's a number of philosophical stances that are taken by this project and by philosophical stance I mean uh practical pragmatically Justified perspectives a way of seeing in the world first I look at active inference as a normative framework I don't think that's controversial at all I mean that's that's what it's defined to be and uh to persist an agent must behave as if it acts to minimize surprisal now clearly I'm not implementing the uh Society of mine using the mathematics of the free energy principle but I'm implementing it hopefully such that doing an active inference analysis of the robots observable behaviors in states will demonstrate that the uh Society of Mind the agent as a whole is acting such as to reduce surprisal so I'm implementing an agent and also its environment as a system of generative processes the active inference analysis will be done in terms of generative models that are inferred uh from observing the behaviors of the robot and

uh observing the states that are visible to an outside Observer other uh philosophical stances I'm just going to go through them very quickly each one is worth an entire presentation I I I take the sense that cognition arises from an EMB Ed agent engaging its environment that sense making is the discovery of unified causal theories that meaning is grounded in the agent's drive to survive no mortality no meaning that intelligent emerges from interactions between a collective of parts that's the Society of mind and that the parts constrain the hole and the hole constraints the part so constraint closure I'm implementing I'm in the process of implementing this this this this system that will hopefully demonstrate artificial agency and here's the current progress there are three phases in this project phase one I developed an cognitive architecture that's what I presented today uh implemented a one version of the the app perception engine and that's what I presented last year I implemented the subsystem to uh realize a to connect to sensors and actuators that's the Cara body and I implemented a simulated grid World in which a simulated um agent would would operate and and and interact that's to uh speed up uh implementation because it takes a long time to uh if I were to do this only in the real world the um the the edit run debug Loop would be extremely long and so I'm in the process right now of implementing uh the agency uh capabilities in phase two um which would be about uh six months to a year from now I will have a a redesigned robot body I will um uh be testing in a real life environment and I will be collecting a lot of data through a a The Observer subsystem and in the third and final phase the uh data gathered from runs both in real life and in the virtual world will be subjected to active inference analysis to validate that my agent does survive because it minimizes Sur risal that's it um if you are interested you can contact me at uh this email address JF do c Lou T ractive inf. Institute you can also uh reach me on uh the Sim Cog robotic channel of the institutes Discord and if you're interested in the source code it is on GitHub so I'd like to uh thank uh the active INF Institute for uh support and encouragement